

EquiOrient: Evaluating Latent Transformation Equivariance for Compositional Spatial Reasoning

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Abstract

001 Can training a VLM’s answer-relevant spatial latent to
002 obey a predeclared geometric transformation algebra
003 improve compositional generalization beyond matched
004 augmentation and output-consistency baselines? We
005 test this on the dihedral group D_4 (horizontal reflection
006 H and 90° rotation R , with five held-out compositions
007 including noncommutative pairs like RH), using 8-
008 way compass-direction labels on synthetic scenes with
009 dense homogeneous clutter. We compare six matched
010 arms differing only in structural objective, under a
011 fully deterministic protocol (seeded initialization, SHA-
012 256-seeded data noise, provenance-annotated results).
013 EquiOrient achieves strong, specific latent algebra com-
014 pliance: the structural loss is nonzero and convergent,
015 and the wrong-geometry arm learns its own incorrect
016 law. Behaviorally, at the scarce-data regime ($N=128$,
017 fifteen matched seeds) behavior is sensitive to the cor-
018 rectness of the imposed law: the correct law is signifi-
019 cantly distinguishable from the wrong law (+0.68pp,
020 95% CI [+0.5, +0.9]pp, $p < 0.0001$, 15/15 seeds),
021 yet correct structural supervision does not outperform
022 plain augmentation (−0.9pp, CI [−2.3, +0.5]pp, n.s.).
023 At $N \geq 512$ every arm reaches ceiling (≥ 0.999) be-
024 cause the VLM backbone’s region-pooled features are
025 already robust to D_4 transforms. The result is a pre-
026 cisely quantified mixed outcome: at low data, behav-
027 ior is sensitive to whether the imposed geometric law
028 is correct, but correct structural supervision does not
029 outperform unconstrained augmentation, and no be-
030 havioral differentiation survives once the backbone sat-
031 urates the task.

032 1. Introduction

033 Spatial reasoning in vision–language models (VLMs)
034 often requires compositional generalization: predict-
035 ing properties of object relationships under transfor-

mations not seen during training. A natural hypothesis
is that forcing a model’s latent representation to obey
the correct transformation algebra should improve this
generalization. We test this hypothesis rigorously.

Phase 1 of EquiOrient [?] established that a
manipulation-verified equivariance objective can train
a VLM’s answer-path latent to obey a predeclared
transformation algebra with strong specificity. How-
ever, that evaluation had critical limitations: a binary
left/right task where the answer algebra is closed, one
seed, and a composition test ($V \circ H$) where the wrong
law coincidentally matched the correct one.

This paper presents Phase 2, which fixes every iden-
tified limitation:

- Noncommutative group: D_4 generated by H and R ,
where $RH \neq HR$, eliminating the Phase 1 symme-
try collision.
- 8-way directional labels: Eight compass directions,
giving output consistency a legitimate non-trivial
permutation.
- Independent scenes: 4,096 unique scenes with 12–20
homogeneous distractors, split by scene ID with no
leakage.
- Five seeds $\times$ six arms: 30 confirmatory runs for tight
confidence intervals.
- Identifiability audit: Pre-GPU verification that
held-out group elements distinguish correct from
wrong representations.
- Collapse-resistant metrics: Normalized equivari-
ance error, effective rank, cosine agreement, and z -
ablation tests.

Our central finding is a precisely quantified mixed
outcome. EquiOrient’s structural objective works at
the representation level: the latent obeys the correct
algebra and is behaviorally distinguishable from an arm
trained with an incorrect law even in the scarce-data
regime ($N=128$, +0.68pp across fifteen matched seeds,
 $p < 0.0001$). However, correct structural supervision
does not outperform plain augmentation at low data

075 (-0.9pp, n.s.) , and at $N \geq 512$ all arms reach ceiling
076 because the VLM backbone’s region-pooled features
077 are already D_4 -robust. For modern VLMs on this syn-
078 thetic plan-view task, behavior is sensitive to whether
079 the imposed geometric law is correct, but explicit ge-
080 ometric structure provides no measurable benefit over
081 unconstrained training once the backbone saturates the
082 task.

083 2. Related Work

084 Latent group representations. The Homomorphism
085 AutoEncoder [4] learns group-structured latents from
086 observed transitions. EquiOrient’s group action is
087 predeclared, not learned, and the question is VLM-
088 specific.

089 Representation-level geometric supervision.
090 GASP [3] injects 3D spatial priors into VLMs
091 via correspondence invariance. EquiOrient asks
092 whether relation-relevant components should trans-
093 form non-uniformly, with a wrong-action causal
094 control.

095 Transformation-pair consistency. SAGE [5] trains
096 VLMs for logical consistency under paired operations.
097 EquiOrient treats output consistency as a matched
098 baseline and adds the representation-level structural
099 objective.

100 Equivariant representation learning. Cohen &
101 Welling [2] and Bronstein et al. [1] are foundational.
102 EquiOrient’s contribution is the VLM answer-path
103 instantiation with a typed heterogeneous algebra and
104 matched controls.

105 3. Method

106 3.1. D_4 algebra and representation

107 The dihedral group D_4 is generated by H (horizontal
108 reflection) and R (90° CCW rotation) with relations
109 $H^2 = R^4 = I$ and $HRH = R^{-1}$. In image coordinates
110 (x rightward, y upward):

$$111 \quad H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

112 The eight elements are
113 $\{I, R, R^2, R^3, H, RH, R^2H, R^3H\}$. Training ex-
114 poses only I , H , and R ; the five elements
115 $\{R^2, R^3, RH, R^2H, R^3H\}$ are held-out.

116 The latent is $\mathbf{z} = [\mathbf{z}_x; \mathbf{z}_y] \in \mathbb{R}^{256}$ with $\mathbf{z}_x, \mathbf{z}_y \in \mathbb{R}^{128}$,
117 produced by a pair encoder on region-pooled Qwen3-
118 VL-8B deepstack features. The correct representation

is $\rho(g) = G_g \otimes I_{128}$, giving:

$$\rho(H)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_x, \mathbf{z}_y], \quad \rho(R)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_y, \mathbf{z}_x]. \quad 120$$

The wrong representation uses $\tilde{\rho}(R) = R_{180} = -I$
and $\tilde{\rho}(H) = H$. This is algebraically self-consistent
but geometrically incorrect; crucially, $\tilde{\rho}(R^2) = I$ while
 $\rho(R^2) = -I$, so the held-out set distinguishes correct
from wrong.

3.2. Identifiability audit

Before any GPU run, we verify on all eight group
elements that $\rho(g) \neq \tilde{\rho}(g)$ for every held-out ele-
ment. The collision matrix confirms zero collisions on
 $\{R^2, R^3, RH, R^2H, R^3H\}$, guaranteeing the evaluation
can in principle distinguish the two laws.

3.3. Eight-way directional task

Each scene contains one target pair (a, b) with displace-
ment $\Delta = (x_a - x_b, y_a - y_b)$. The label is the compass
direction of Δ among eight classes: right, upper-right,
above, upper-left, left, lower-left, below, lower-right.
The exact label permutation π_g under each $g \in D_4$ is
computed from the group action on Δ .

3.4. Six matched arms

All arms share: Qwen3-VL-8B-Instruct (frozen
text backbone), vision LoRA (rank 16,
qkv/proj/c_fc/c_proj), PairEncoderV2 (8192→512
→256), RelationHead8 (256→8), AdamW (lr 10^{-4} ,
batch 8, 2 epochs). The only difference is the
structural objective:

Arm	Answer loss	Structural term
Original SFT	identity only	—
Augmentation	identity + transform	—
Output consistency	identity + transform	$\text{MSE}(\ell_{Tx}, \kappa_g(\ell_x))$
Latent invariance	identity + transform	$\ \mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
EquiOrient	identity + transform	$\ \rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
Wrong geometry	identity + transform	$\ \tilde{\rho}(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$

Sparse exposure: each training scene contributes
identity + exactly one generator (H or R , balanced
50/50). A manipulation check asserts structural loss
> 10^{-6} every epoch.

3.5. Dataset

4,096 independent scenes (512 dev, 2048 train pool,
512 val, 1024 test). Each scene: one target pair (2-
4px), 12–20 homogeneous distractors (3-color muted
palette), low-contrast background, per-pixel noise (am-
plitude 12). Balanced 8-way labels. Scene IDs disjoint
across splits. Difficulty was escalated across four ver-
sions (v1–v4); all four produced behavioral ceiling.

Arm	Unseen acc.	Struct. loss
Original SFT	0.9994 ± 0.0007	0.000
Augmentation	1.0000 ± 0.0000	0.000
Output consistency	0.9992 ± 0.0014	0.459
Latent invariance	0.9998 ± 0.0004	0.224
EquiOrient	1.0000 ± 0.0000	0.112
Wrong geometry	1.0000 ± 0.0000	0.176

Table 1. Unseen-group accuracy and structural loss (mean $\pm$ std over five seeds). All arms are at ceiling.

Arm	I	R	R2	R3	H	RH	R2H	R3H
Aug.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Eq.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Wrong	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Inv.	0.999	1.000	1.000	1.000	0.999	0.999	1.000	1.000
SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999	0.999

Table 2. Per-transform accuracy (5-seed mean). All arms at ceiling.

N	Aug.	Eq.	Wrong	Eq-Aug (CI)
128	0.955 ± 0.021	0.945 ± 0.016	0.939 ± 0.017	$-0.9\text{pp} [-2.3, +0.5]$
512	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.0pp
2048	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.0pp

Table 3. Unseen-group accuracy (mean $\pm$ std). $N=128$: 15 matched deterministic seeds. $N=512$: 30 runs. $N=2048$: 20 runs.

higher structural loss (0.459) because its MSE objective operates on logit space rather than the structured latent.

Training dynamics. EquiOrient’s answer loss converges from 4.106 $\rightarrow$ 0.211 across two epochs, while its structural loss decreases from 0.318 $\rightarrow$ 0.108.

The wrong-geometry arm shows analogous convergence (4.424 $\rightarrow$ 0.388 answer, 0.351 $\rightarrow$ 0.174 structural). Both manipulations checks pass: structural losses are nonzero and decreasing.

4. Verification Protocol

We enforce a 45-test pre-GPU gate covering: D_4 group axioms (4 tests), label action correctness (3), renderer correctness (4), dataset integrity (5), structural loss properties (4), gradient flow (5), matched-architecture invariants (3), collapse checks (4), and identifiability (2). All 45 must pass before any GPU run. This suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

5. Results

5.1. Confirmatory results (5 seeds)

1 reports the confirmatory results across five seeds ($N = 512$, $\lambda = 1.0$).

Behavioral ceiling. Every arm achieves ≥ 0.999 unseen-group accuracy with overlapping confidence intervals. Per-transform accuracy is ≥ 0.998 for all eight D_4 elements across all arms (Table 2). The hierarchical paired bootstrap yields $\Delta A = A_{\text{EquiOrient}} - A_{\text{augmentation}} = 0.0000$ with 95% CI [0.0000, 0.0000], ruling out a $\geq 3\text{pp}$ benefit.

Latent equivariance. EquiOrient’s structural loss is 0.112 ± 0.005 (epoch 2), confirming the latent is constrained to obey the correct algebra. The wrong-geometry arm’s structural loss is 0.176 ± 0.010 —it learns its own (incorrect) law. Output consistency has

Why the ceiling exists. The bounding-box pooling mechanism provides ground-truth object positions to the model. The VLM backbone’s deepstack features encode relative object positions in a way that is inherently robust to global D_4 transforms. Making objects smaller, adding more distractors, reducing color diversity, increasing noise—none of these break the ceiling, because the model never needs to find the objects. The backbone’s region-pooled spatial features are already D_4 -equivariant by construction.

5.2. Sample efficiency (data-scale experiments)

To test whether the structural objective provides benefit when data is scarce, we trained four key arms (augmentation, output consistency, EquiOrient, wrong geometry) at $N = 128$ training scenes with fifteen matched seeds under a fully deterministic protocol (all random seeds fixed before model initialization; dataset noise seeded via SHA-256 of scene ID and transform, identical across processes). We additionally report $N = 512$ (30 runs) and $N = 2048$ (20 runs).

At $N=128$, no arm reaches ceiling. EquiOrient’s paired contrast against augmentation is -0.9pp with 95% CI $[-2.3, +0.5]\text{pp}$ (paired t -test $p = 0.22$, wins 5/15 seeds): there is no evidence that the correct geometric structure improves behavior beyond plain augmentation. However, the predeclared contrast against the wrong-geometry control is $+0.68\text{pp}$ with 95% CI $[+0.5, +0.9]\text{pp}$ ($p < 0.0001$, wins 15/15 seeds): the correct law is behaviorally distinguishable from the incorrect one even at $N=128$. At $N \geq 512$, all arms reach

225 ceiling and every contrast is identically zero.

226 The picture is therefore mixed and precisely quanti-
227 fied: at low data, behavior is sensitive to whether the
228 imposed geometric law is correct (the correct law reli-
229 ably outscores the wrong law), but correct structural
230 supervision does not outperform an unconstrained aug-
231 mentation baseline, and all behavioral differences van-
232 ish once the backbone saturates the task.

233 6. Discussion

234 The structural objective works; the task does not.
235 EquiOrient successfully trains the latent to obey the
236 correct algebra: the structural loss is nonzero and con-
237 vergent ($0.318 \rightarrow 0.108$), and the wrong-geometry arm
238 independently learns its own incorrect law ($0.351 \rightarrow$
239 0.174). The mechanism works. However, behav-
240 ioral accuracy is at ceiling for all arms, because the
241 bounding-box pooling provides ground-truth object
242 positions to the model, eliminating the perceptual chal-
243 lenge that spatial compositionality requires.

244 Why visual difficulty does not help. We systemati-
245 cally escalated difficulty across four versions (v1-v4):
246 reducing target size ($6\text{-}9\text{px} \rightarrow 2\text{-}4\text{px}$), increasing dis-
247 tractor count ($6\text{-}10 \rightarrow 12\text{-}20$), reducing color diver-
248 sity ($12 \rightarrow 3$ near-identical hues), and increasing per-
249 pixel noise (amplitude $6 \rightarrow 12$). All four versions pro-
250 duced behavioral ceiling. The cause is architectural:
251 the model pools features from grid cells specified by
252 bounding boxes, so it never needs to locate objects or
253 reason about their visual properties. The backbone’s
254 region-pooled spatial features encode relative positions
255 in a way that is inherently robust to global D_4 trans-
256 forms.

257 Implications for compositionality benchmarks. Our
258 result reveals a design pitfall: benchmarks that pro-
259 vide bounding boxes as input may systematically over-
260 estimate spatial compositionality difficulty for modern
261 VLMs. The backbone’s visual features already encode
262 the needed geometric information. Future benchmarks
263 should consider: (1) natural images without bound-
264 ing boxes, where object localization is part of the chal-
265 lenge; (2) relations requiring multi-step reasoning be-
266 yond single-pair displacement; (3) adversarial construc-
267 tion targeting known VLM failure modes; and (4) tasks
268 where the answer cannot be inferred from a single pair’s
269 pixel coordinates.

270 What this paper contributes. The contribution is a
271 precisely quantified mixed result with an architectural

272 explanation. At scarce data ($N=128$, ten matched de-
273 terministic seeds), the correct geometric law is behav-
274 iorally distinguishable from the incorrect one ($+0.76\text{pp}$,
275 $p < 0.001$, all ten seeds) but not from plain augmen-
276 tation (-1.4pp , n.s.): structural constraints encode
277 which law the latent obeys, yet provide no measur-
278 able advantage over an unconstrained baseline. At
279 $N \geq 512$, bounding-box-pooled VLM features are al-
280 ready D_4 -robust, so every arm reaches ceiling and all
281 contrasts vanish. The finding is actionable: it tells the
282 field that providing bounding boxes as input shortcuts
283 the spatial reasoning challenge, and that future work
284 on geometric structure in VLMs must evaluate without
285 such shortcuts to test whether the correct law confers
286 a genuine behavioral advantage.

517 7. Limitations

288 This study has several limitations that should guide
289 future work:

- 290 • Bounding-box shortcut. The model receives ground-
291 truth object positions, eliminating the perceptual
292 challenge. The finding that behavioral accuracy is
293 at ceiling is an artifact of this design choice, not a
294 general statement about VLM spatial reasoning.
- 295 • Frozen backbone. The VLM text backbone is
296 frozen; only the vision LoRA and downstream heads
297 are trained. The structural objective cannot modify
298 the backbone’s spatial representations. Future work
299 should test with an unfrozen backbone.
- 300 • Synthetic data. All scenes are synthetic plan-view
301 renderings. The finding that VLM features are D_4 -
302 robust may not transfer to natural images where
303 rotations introduce semantic artifacts.
- 304 • Two epochs of training. Each arm trains for only
305 2 epochs (~ 128 steps). Longer training could re-
306 veal behavioral differences between arms, though
307 the ceiling suggests otherwise.
- 308 • Single group. Only D_4 (8 elements) is tested. Larger
309 groups ($\text{SO}(2)$, $\text{SE}(3)$) may produce harder tasks
310 where structural objectives provide benefit.
- 311 • No comparison to prior methods. SAGE [5] and
312 GASP [3] are cited but not compared head-to-head.

313 8. Conclusion

314 We present EquiOrient Phase 2, a rigorous evaluation
315 of latent transformation equivariance on the noncom-
316 mutative dihedral group D_4 , with 8-way labels, 30
317 confirmatory runs at $N=512$, ten matched determin-
318 istic seeds at $N=128$, and a 45-test verification suite.
319 EquiOrient achieves strong, specific latent algebra com-
320 pliance (structural loss 0.112, wrong-geometry 0.176).
321 Behaviorally, the correct law is significantly distin-

guishable from the wrong law at $N=128$ ($+0.76pp$, $p < 0.001$) but not from plain augmentation ($-1.4pp$, n.s.), and at $N \geq 512$ all arms reach ceiling because bounding-box-pooled VLM features are already D_4 -robust. The result is a precisely quantified mixed outcome with an architectural explanation: geometric structure encodes which law the latent obeys, but provides no measurable advantage once the backbone saturates the task. Future work must evaluate without bounding-box shortcuts to meaningfully test whether correct geometric structure confers a genuine behavioral advantage.

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